

Pranav Kalakota

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Education

Purdue University

Bachelor of Science in Computer Science

West Lafayette, IN

Aug. 2025 – May 2029

- Coursework: Programming in C, Object-Oriented Programming, Linear Algebra, Discrete Mathematics, Statistical Methods

Experience

Data Science Researcher

Johnson & Johnson

Aug. 2026 – Present

West Lafayette, IN

- Building a Copilot Studio agent over Power Automate cloud flows to automate JJ's 45-minute, 10-step booking process, using Microsoft Graph for calendar availability and attendee resolution.
- Designing a multi-agent Copilot Studio architecture within an 8,000-character context window, compressing instructions into Word doc references to reclaim 2,000+ characters for scheduling logic.
- Sandboxing JJ's Microsoft environment across a 12-person team, using Power BI to benchmark prompt reliability and reduce Copilot's 10% template-adherence failure rate.

Software Engineering Intern

Crcle

June 2026 – Present

West Lafayette, IN

- Architected a multimodal RAG pipeline fusing Whisper ASR, Llama 3.2 3B, LLaVA 7B, ArcFace, and ECAPA-TDNN with cosine-similarity gating over ChromaDB ONNX vector store, enabling zero-touch HCI via cross-app voice control with sub-20s latency and zero cloud dependency.
- Engineered a schema-constrained NLP intent router with 3-tier fuzzy/regex/LLM fallback and semantic vector search, achieving 95% deterministic routing across 500+ fuzz invariants and 12+ query types in production.
- Optimized always-on wearable ASR pipeline with adaptive windowing and CPU-routed Whisper inference, cutting transcription latency 50% (12s to 6s) and wake-word activation 90% (2s to <200ms) via offline-first ONNX detection with parallel ASR fallback.
- Built a cross-platform vision-based GUI automation system achieving sub-2s latency and 90%+ element localization accuracy across arbitrary applications by routing screenshots through Gemini multimodal LLM inference with pixel-precise coordinate extraction.

Software Engineering Intern

Pololu Robotics

June 2026 – Aug. 2026

Las Vegas, NV

- Engineered an automated production test system in Arduino with serial CLI and fault handling, validating dual H-bridge MOSFET output, current thresholds, and fault flags across 15+ ADC channels, cutting QA cycle time by 10%.
- Refactored 500+ line legacy embedded codebase, resolved a critical timing race condition in PWM chopping logic, and validated power-on-reset sequencing for long-term maintainability.

Projects

LSM-Tree Storage Engine | Java

Present

- Building an LSM-tree storage engine with MemTable, write-ahead log, SST files, bloom filters, and multi-level compaction targeting 50K+ writes/sec.
- Implementing bloom filter-gated SST reads and background compaction targeting 3x read speedup and sub-5ms p99 latency across 100K+ keys.

MCP Task Manager | TypeScript, SQLite, Express, MCP Protocol

Jan. 2026 – Mar. 2026

- Engineered a dual-architecture task management system exposing 5 tools over MCP via stdio transport with an Express web dashboard backed by a shared SQLite database.
- Designed a graceful degradation layer falling back from OpenAI API to a local regex-based parser with fuzzy typo tolerance, ensuring zero downtime during API failures.

Honors & Awards

Claude Builder Hackathon: 1st place in the healthcare track for VEDA

Databricks Generative AI Fundamentals

Deloitte Data Analytics Certificate

Technical Skills

Languages: Java, Python, C, Swift, TypeScript, HTML/CSS

Tools & Platforms: Git, Arduino, SolidWorks, ONNX Runtime, ChromaDB, Power Automate, Jupyter Notebook, Tableau

Libraries & Frameworks: Matplotlib, Pandas, Express, Whisper, LLaVA, Vision Framework

Concepts: Object-Oriented Programming, Databases, Simulations, Data Analysis, Embedded Systems, RAG Pipelines